Curriculum Standard for Information Technology

Career Cluster: Web and Digital Communications, Information Support and Services, Programming and Software Development, and Network Systems.

Cluster Description: Building linkages in IT occupations framework: for entry level, technical, and professional careers related to the design, development, support, and management of hardware, software, multi-media, and systems integration services.

Pathway: Information Technology | **Effective Term:** Fall 2025 (2025*03)

Program Major Under Pathway

Program Major / Classification of Instruction Programs (CIP) Code		Credential Level(s) Offered	Program Major Code
Information Technology	CIP Code: 11.0103	AAS/Diploma/Certificate	A25590

Pathway Description:

The Information Technology (IT) curriculum prepares graduates for employment in the technology sector as designers, testers, support technicians, system administrators, developers, or programmers who use computer software and\or hardware to design, process, implement and manage information systems in specialties such as database services, security, business intelligence, healthcare informatics and others depending on the technical path selected within this curriculum.

Course work includes development of a student's ability to create, store, communicate, exchange and use information to solve technical issues related to information support and services, interactive media, network systems, programming and software development, information security and other emerging technologies based on the selected area of study.

Graduates should qualify for employment in entry-level positions with businesses, educational systems, and governmental agencies which rely on computer systems to design and manage information. The program will incorporate the competencies of industry-recognized certification exams.

I. General Education Academic Core

Curriculum Requirements for associate degree, diploma, and certificate programs in accordance with 1 D SBCC 400.10: *Degree programs must contain a minimum of 15 semester hours with a minimum of 6 semester hours of communications (3 SHC of which must be ENG-110 or ENG-111) and the remaining hours from each of the following areas: humanities/fine arts, social/behavioral sciences, and natural sciences/mathematics. Diploma programs must contain a minimum of 6 semester hours of general education, including at least 3 semester hours in communications and the remaining hours from one of the following areas: humanities/fine arts, social/behavioral sciences, and natural sciences/mathematics. General education is optional in certificate programs.*

Information Technology

Recommended General Education Academic Core				AAS	Diploma	Certificate
Minimum General Education Hours Required:				**15 SHC**	**6 SHC**	**0 SHC**
Courses listed below are recommended general education courses for this curriculum standard. Colleges may choose to include additional or alternative general education courses to meet local curriculum needs.						
Communication:						
COM	110	Introduction to Communication	3 SHC	**6 SHC**	**3 SHC**	**Optional**
COM	120	Intro Interpersonal Com	3 SHC			
COM	231	Public Speaking **U**	3 SHC			
ENG	111	Writing and Inquiry **U**	3 SHC			
ENG	112	Writing/Research in the Disc **U**	3 SHC			
ENG	114	Prof Research & Reporting	3 SHC			
ENG	116	Technical Report Writing	3 SHC			
Humanities/Fine Arts:				**3 SHC**	**0-3 SHC**	**Optional**
ART	111	Art Appreciation **U**	3 SHC			
HUM	110	Technology and Society	3 SHC			
HUM	115	Critical Thinking	3 SHC			
HUM	230	Leadership Development	3 SHC			
MUS	110	Music Appreciation **U**	3 SHC			
PHI	240	Introduction to Ethics **U**	3 SHC			

**Within the degree program, the institution shall include opportunities for the achievement of competence in reading, writing, oral communication, fundamental mathematical skills, and basic use of computers.*

Social/Behavioral Sciences:				**3 SHC**	**0-3 SHC**	**Optional**
ECO	151	Survey of Economics	3 SHC			
ECO	251	Prin of Microeconomics **U**	3 SHC			
ECO	252	Prin of Macroeconomics **U**	3 SHC			
POL	120	American Government **U**	3 SHC			
PSY	118	Interpersonal Psychology	3 SHC			
PSY	150	General Psychology **U**	3 SHC			
SOC	210	Introduction to Sociology **U**	3 SHC			
Natural Sciences/Mathematics:				**3-4 SHC**	**0-3 SHC**	**Optional**
MAT	110	Math Measurement and Literacy	3 SHC			
MAT	121	Algebra/Trigonometry I	3 SHC			
MAT	143	Quantitative Literacy **U**	3 SHC			
MAT	152	Statistical Methods I **U**	4 SHC			
MAT	171	Precalculus Algebra **U**	4 SHC			
U indicates a Universal General Education Transfer Component (UGETC) course included in the Comprehensive Articulation Agreement. UGETC courses are guaranteed to transfer to any of the sixteen University of North Carolina senior institutions as equivalent credit within defined distribution limits.						

II. Major Hours. AAS, diploma, and certificate programs must include courses that offer specific job knowledge and skills. Work Based Learning (WBL) may be included in associate in applied science degrees up to a maximum of 8 semester hours of credit; in diploma programs up to a maximum of 4 semester hours of credit; and in certificate programs up to a maximum of 2 semester hours of credit. Below is a description of each section under Major Hours.

A. Technical Core. The technical core is comprised of specific courses which are required for all Program Majors under this Curriculum Standard. A diploma program offered under an approved AAS program standard or a certificate, which is the highest credential level awarded under an approved AAS program standard, must include a minimum of 12 semester hours of credit derived from the curriculum core courses or core subject area of the AAS program.

B. Program Major(s). The Program Major must include a minimum of 12 semester hours of credit from required subjects and/or courses.

C. Other Major Hours. Other Major Hours must be selected from prefixes listed on the curriculum standard. A maximum of 9 semester hours of credit may be selected from each prefix listed, with the exception of prefixes listed in the core.

Information Technology				**AAS**	**Diploma**	**Certificate**
Minimum Major Hours Required:				**49 SHC**	**30 SHC**	**12 SHC**
A. Technical Core:				**18-25 SHC**	**12 SHC**	
A diploma offered under this AAS degree requires a minimum of 12 SHC extracted from the required subject/course core of the AAS degree						
CTS	115	Info Sys Business Concepts	3 SHC			
CTI	110	IT Foundations	3 SHC			
CTI	120	Network & Sec Foundations	3 SHC			
Select one:						
CIS	110	Introduction to Computers	3 SHC			
CIS	115	Intro to Prog and Logic	3 SHC			
CSC	120	Computing Fundamentals I	4 SHC			
CTI	115	Computer Systems Foundation	3 SHC			
CTS	120	Hardware/Software Support	3 SHC			
NOS	110	Operating Systems Concepts	3 SHC			
OST	137	Office Applications I	3 SHC			
WEB	110	Web Development Fundamentals	3 SHC			
Information Technology **Required Subject Areas (Select one subject area):**						
Information Systems *12 SHC to be selected from:* CIS, CSC, CTI, CTS, DME, NOS and/or WEB prefixes						

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Artificial Intelligence						
	CSC	113	Artificial Intel. Fundamentals	3 SHC		
Select one:						
	CSC	112	Machine Learning Computation	3 SHC		
	CSC	114	Artificial Intelligence I	3 SHC		
	CSC	115	Machine Learning I	3 SHC		
Cloud Computing and Virtualization						
Select one:						
	CTI	140	Virtualization Concepts	3 SHC		
	CTI	141	Cloud & Storage Concepts	3 SHC		
	CTI	240	Virtualization Admin I	3 SHC		
Select one:						
	CTI	241	Virtualization Admin II	3 SHC		
	NOS	120	Linux Single User	3 SHC		
	NOS	220	Linux Admin I	3 SHC		
	NOS	230	Windows Administration I	3 SHC		
Computer Programming and Development						
Select one:						
	CSC	121	Python Programming	3 SHC		
	CSC	134	C++ Programming	3 SHC		
	CSC	139	Visual BASIC Programming	3 SHC		
	CSC	151	JAVA Programming	3 SHC		
	CSC	153	C# Programming	3 SHC		
Select one:						
	CSC	221	Advanced Python Programming	3 SHC		
	CSC	234	Advanced C++ Programming	3 SHC		
	CSC	249	Data Structure & Algorithms	3 SHC		
	CSC	251	Advanced JAVA Programming	3 SHC		
	CSC	253	Advanced C# Programming	3 SHC		
Cybersecurity						
Select one:						
	CCT	110	Intro to Cyber Crime	3 SHC		
	CCT	112	Ethics & High Technology	3 SHC		
	CCT	240	Data Recovery Techniques	3 SHC		
	CCT	250	Network Vulnerabilities I	3 SHC		
Select one:						
	SEC	151	Intro to Protocol Analysis	3 SHC		
	SEC	160	Security Administration I	3 SHC		
	SEC	210	Intrusion Detection	3 SHC		
Database Administration and Data Center Technologies						
Select two:						
	BAS	120	Intro to Analytics	3 SHC		
	DBA	120	Database Programming I	3 SHC		
	DBA	210	Database Administration	3 SHC		
	CTI	140	Virtualization Concepts	3 SHC		
	CTI	141	Cloud & Storage Concepts	3 SHC		
	NOS	120	Linux Single User	3 SHC		
Data Management						
Select two:						
	BUS	228	Business Statistics	3 SHC		
	CTS	130	Spreadsheet	3 SHC		
	DBA	110	Database Concepts	3 SHC		
	DBA	115	Database Applications	3 SHC		
	DBA	120	Database Programming I	3 SHC		

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Data Science and Analytics						
Select two:						
BAS	120	Intro to Analytics	3 SHC			
DBA	110	Database Concepts	3 SHC			
DBA	120	Database Programming I	3 SHC			
CSC	121	Python Programming	3 SHC			
CSC	123	R Programming	3 SHC			
CSC	152	SAS	3 SHC			
Digital Media Production						
Select two:						
DME	110	Intro to Digital Media	3 SHC			
DME	115	Digital Media Tools	3 SHC			
DME	120	Intro to Multimedia	3 SHC			
DME	130	Digital Animation I	3 SHC			
WEB	210	Web Design	3 SHC			
Game and Simulation Programming						
Select one:						
CSC	134	C++ Programming	3 SHC			
CSC	151	JAVA Programming	3 SHC			
CSC	153	C# Programming	3 SHC			
SGD	113	SGD Programming I	3 SHC			
Select one:						
SGD	111	Introduction to SGD	3 SHC			
SGD	112	SGD Design I	3 SHC			
IT Business Support						
Select two:						
CTS	130	Spreadsheet	3 SHC			
CTS	240	Project Management	3 SHC			
DBA	110	Database Concepts	3 SHC			
DBA	240	Database Analysis/Design	3 SHC			
WEB	151	Mobile Application Dev I	3 SHC			
IT Project Management						
Select two:						
CSC	114	Artificial Intelligence I	3 SHC			
CTS	240	Project Management	3 SHC			
PMT	110	Intro to Project Mgmt	3 SHC			
PMT	111	Proj Mgmt Assessing Risk	3 SHC			
Mobile Applications Developer						
Select two:						
CTI	150	Mobile Computing Devices	3 SHC			
WEB	125	Responsive Web Frameworks	3 SHC			
WEB	141	Mobile Interface Design	3 SHC			
WEB	151	Mobile Application Dev I	3 SHC			
WEB	251	Mobile Application Dev II	3 SHC			
Network Management						
Select one:						
NET	110	Networking Concepts	3 SHC			
NET	125	Introduction to Networks	3 SHC			
Select one:						
CTI	140	Virtualization Concepts	3 SHC			
CTI	141	Cloud & Storage Concepts	3 SHC			
NET	126	Switching and Routing	3 SHC			
NOS	120	Linux Single User	3 SHC			
NOS	230	Windows Administration I	3 SHC			
TNE	111	Campus Networks I	3 SHC			

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Software and Web Development						
Select two:						
CSC	139	Visual BASIC Programming	3 SHC			
CSC	143	Object-Oriented Prog	3 SHC			
CSC	151	JAVA Programming	3 SHC			
CSC	153	C# Programming	3 SHC			
DBA	120	Database Programming I	3 SHC			
WEB	115	Web Markup and Scripting	3 SHC			
Support and Services						
CTS	155	Tech Support Functions	3 SHC			
Select one:						
CET	211	Computer Upgrade/Repair II	3 SHC			
CTS	135	Integrated Software Intro	4 SHC			
CTS	220	Adv Hard/Software Support	3 SHC			
CTS	250	User Support & Software Eval	3 SHC			
CTS	255	Adv Tech Support Functions	3 SHC			
CTS	272	Desktop Support: Apps	3 SHC			
Systems Security						
SEC	110	Security Concepts	3 SHC			
Select one:						
CCT	110	Intro to Cyber Crime	3 SHC			
NET	110	Networking Concepts	3 SHC			
NET	125	Introduction to Networks	3 SHC			
SEC	150	Secure Communications	3 SHC			
SEC	160	Security Administration I	3 SHC			
Web Administration and Design						
Select two:						
DBA	120	Database Programming I	3 SHC			
NOS	120	Linux Single User	3 SHC			
NOS	230	Windows Administration I	3 SHC			
WEB	115	Web Markup and Scripting	3 SHC			
WEB	210	Web Design	3 SHC			
WEB	250	Database Driven Websites	3 SHC			

C. Other Major Hours.

To be selected from the following prefixes:

ACC, ART, ATR, BAS, BUS, CCT, CET, CIS, CJC, CSC, CTI, CTS, DBA, DEA, DES, DFT, DME, ECM, ECO, EDU, EGR, ELC, ELN, ETR, GIS, GRA, GRD, HBI, HIT, HMT, HUM, ISC, ITN, MAT, MCO, MED, MIT, MKT, NET, NOS, OMT, OST, PCI, PHI, PHO, PMT, SEC, SGD, SGR, TNE, WBL, and WEB.

Up to two semester hour credits may be selected from ACA.

Three semester hour credits may be selected from PTE.

Up to three semester hour credits may be selected from the following prefixes: ARA, ASL, CHI, FRE, GER, IRI, ITA, JPN, LAT, POR, RUS, and SPA.

III. Other Required Hours

A college may include courses to meet graduation or local employer requirements in a certificate (0-1 SHC), diploma (0-4 SHC), or an associate in applied science (0-7 SHC) program. These curriculum courses shall be selected from the Combined Course Library and must be approved by the System Office prior to implementation. Restricted, unique, or free elective courses may not be included as Other Required Hours.

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IV. Employability Competencies

Fundamental competencies that address soft skills vital to employability, personal, and professional success are listed below. Colleges are encouraged to integrate these competencies into the curriculum by embedding appropriate student learning outcomes into one or more courses or through alternative methods.

- **A. Interpersonal Skills and Teamwork** – The ability to work effectively with others, especially to analyze situations, establish priorities, and apply resources for solving problems or accomplishing tasks
- **B. Communication** – The ability to effectively exchange ideas and information with others through oral, written, or visual means
- **C. Integrity and Professionalism –** Workplace behaviors that relate to ethical standards, honesty, fairness, respect, responsibility, self-control, criticism and demeanor
- **D. Problem-solving** – The ability to identify problems and potential causes while developing and implementing practical action plans for solutions
- **E. Initiative and Dependability** – Workplace behaviors that relate to seeking out new responsibilities, establishing and meeting goals, completing tasks, following directions, complying with rules, and consistent reliability
- **F. Information processing** – The ability to acquire, evaluate, organize, manage, and interpret information
- **G. Adaptability and Lifelong Learning –** The ability to learn and apply new knowledge and skills and adapt to changing technologies, methods, processes, work environments, organizational structures and management practices
- **H. Entrepreneurship –** The knowledge and skills necessary to create opportunities and develop as an employee or self-employed business owner

***The North Carolina Career Clusters Guide was developed by the North Carolina Department of Public Instruction and the North Carolina Community College system to link the academic and Career and Technical Education programs at the secondary and postsecondary levels to increase student achievement. Additional information about Career Clusters is located at:* http://www.nc-net.info/NC_career_clusters_guide.php *or* http://www.careertech.org/

Summary of Required Semester Hour Credits (SHC) for each credential:

	AAS	Diploma	Certificate
Minimum General Education Hours	15	6	0
Minimum Major Hours	49	30	12
Other Required Hours	0-7	0-4	0-1
Total Semester Hours Credit (SHC)	**64-76**	**36-48**	**12-18**

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